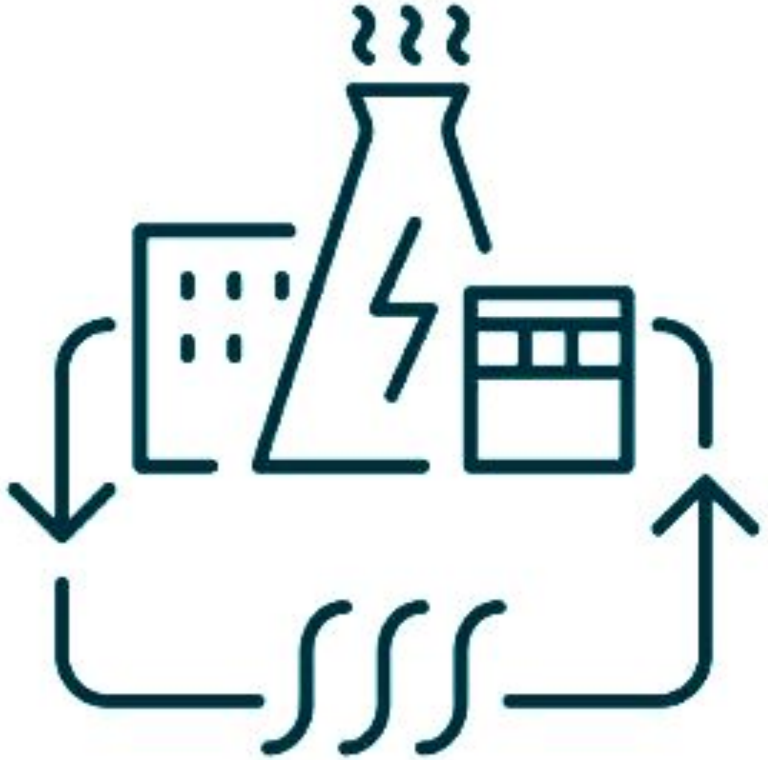
Thermal Energy Network Proposal N.Y.P. and MTA



AGENDA

1 Problem Statement

2 Background

3 Explored Solutions/ Recommendations

4 Technical Elements

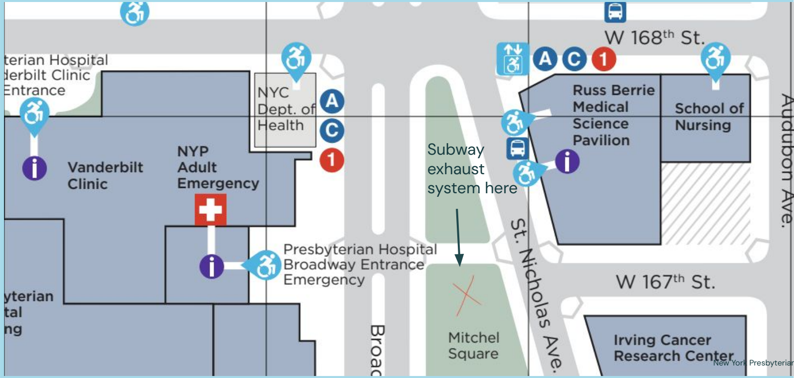
5 Potential Partnerships

6 Cost Benefit Analysis

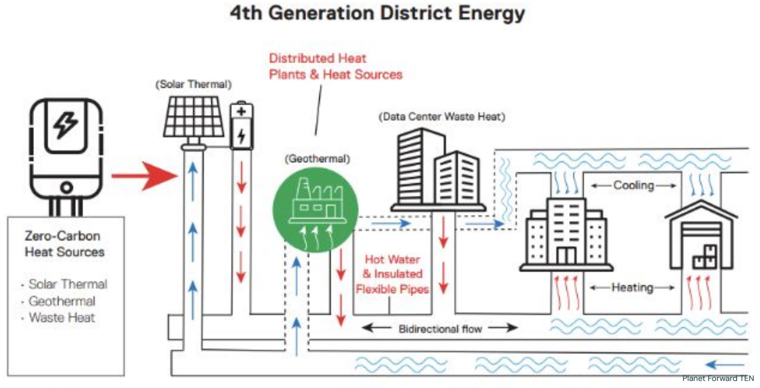
9 Challenges

10 Conclusion

PROBLEM STATEMENT



What is the potential for a chiller-based Thermal Energy Network to capture New York Presbyterian Manhattan Campus waste heat and waste heat from 168th St. station?



Based on Lund et al. (2014); Lund et al. (2021); Sulzer et al. (2021)

BACKGROUND

First example of subway excess heat recapture

Bunhill 2 Energy Center launched 2020, captures subway excess heat and transfers heat to residential buildings

New York Presbyterian

Currently heated by steam boiler system
Committed to reduce carbon emissions by 30% by 2030
Began exploring heat pumps and sustainable heating in 2024

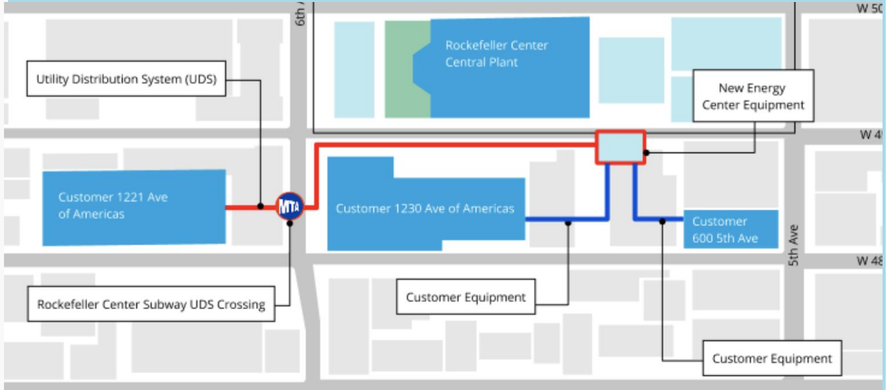
MTA

MTA released RFI 09/18/2025 requesting geothermal cooling technologies for 168th St. station

MTA working with ConEd on TEN for Rockefeller Center

Bunhill

Rockefeller TEN



EXPLORED SOLUTIONS – RECOMMENDATIONS

Solution: Thermal energy network in NYP to utilize chiller waste heat and capture 168th street station excess heat.

01

TEN + HEAT PUMPS

A thermal network connecting the 168th street station to NYP. Informed by case studies that successfully captured waste heat and redistributed through district scale systems.

02

DISTRICT LEVEL EXPANSION

Potential expansion of the network with surrounding buildings and analyzed existing technical instruments to support it.

03

PARTNERSHIPS

Considering NYCHA, Columbia University Irving Center, NYSERDA and other potential partners to help scale the network and increase ROI.

04

COST BENEFIT ANALYSIS

Utilizing the Rockefeller Plaza pilot as a reference to conduct cost benefit analysis on feasibility of a chiller based TEN at NYP.

TECHNICAL ELEMENTS

Feasibility Study

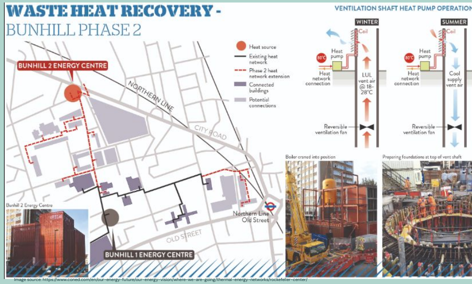
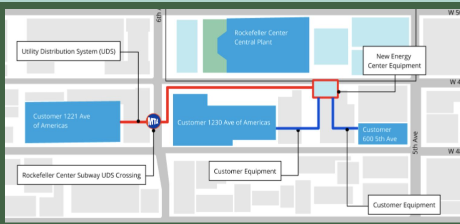
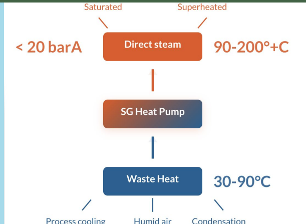
Drawing from existing TENS using subway waste heat	Bunhill 2 in Uk	Captures platform heat and moves through heat pumps, provides heat to more than 1350 homes.	
Con-Ed examples of TENs	Rockefeller TEN Mt. Vernon Chelsea	Recovers waste heat from Rockefeller Center Central Plant traversing through the nearby subway station—to three nearby commercial buildings in Midtown Manhattan.	
Waste Heat to Steam Heat Pumps	Captures low-grade heat and raises it to produce steam using centrifugal compressors	Developed by Heatlift. Generated steam integrates into existing process for drying, heating, evaporation or sterilization and can be recompressed to reach up to 20 barA and more than 200°C.	

Image source: https://www.coned.com/en/our-energy-future/our-energy-vision/where-we-are-going/thermal-energy-networks/rockefeller-center/

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Image source: https://www.heatlift.com/products/hthp/steam-generation-heat-pump/

POTENTIAL PARTNERSHIPS

MTA

Already exploring project feasibility

Columbia University

Partial ownership of campus and invested in decarbonization

New York Presbyterian

Seeking alternative heating options like heat pumps
Assessing feasibility of TEN with MTA

NYCHA

Potential partner for expansion of TEN, Fort Washington Ave. Rehab nearby

NYSERDA

Previously awarded $1.4M to explore energy efficiency measures at NYP

Con-Edison

Piloting three TEN in NYC, Rockefeller Center

COST BENEFIT ANALYSIS/ POTENTIAL COST SCENARIO

$134 million

Total cost of the system **over 5 years**

$7.8 million a year to operate
68,000 MMBTUs Generated

If NYP were to **use** the power generated:

1,110,000 MMBTU – annual demand

6% of energy saved
$285,735.44 annual savings (at $0.042/kBtu)

If NYP were to **sell** the power generated:

$1,394,661.29 in sales at ($0.0205/kBtu)

$0.1156 per kBtu to customers of TEN to cover annual operating costs

Costs required to cover 5 year operation:

$132,797,774.39

if NYP were to **save** the energy

$127,253,145.16

if NYP were to **sell** all the energy at $0.0205/kBtu

$94,887,096.77

if NYP were to **sell** all the energy at $0.1156/kBtu

$1.97

per kBtu required rate to cover costs of total project

MTA IMPACT

529,814.38

kBtu to get rid of

to sell it into TEN / pay the TEN to take the energy

$2,225.22

for the whole summer at $0.0205/kBtu

$61,272.57

for the whole summer at $0.1156/kBtu

CHALLENGES

Cost Benefit

Establishing fair excess heat pricing that justifies cost of installation and retrofits needed to NYP. Question of scale with respect to expanding the network.

Contaminated Air

Hospital has high sanitary requirements. Getting heat from the subway threatens exposure to contaminants from subway excess heat.

Disruption of traffic and trains

Status quo of street traffic and metro function be affected during installation phase.

Waste heat load capture and reliability

The amount of waste heat captured from the subway must justify the project and once implemented must serve as a reliable secondary heat source.

CONCLUSION

Technology exists for thermal Energy Networks can recover subway excess heat to heat surrounding commercial and residential areas like NYP, but it may not be cost effective in this iteration of a TEN.

Thank you

Questions?